

Emin Erkan YILDIRIM

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Education

Dokuz Eylul University – MS in Electrical and Electronics Engineering	February 2025 – Present
Dokuz Eylul University – BS in Electrical and Electronics Engineering (GPA: 3.0/4.0)	September 2019 – June 2024

Experience

Laboratory Specialist R&D , Vestel Beyaz Eşya – Manisa, Türkiye	October 2024 – Present
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- Verified electronic control boards used in tumble dryers to ensure optimal functionality, performance, and compliance with IEC 60335 standards.
- Prepared engineering requirements and technical specification documents.
- Conducted benchmark studies and analyzed competitor products to identify new components suitable for integration.
- Made design decisions to ensure product safety and regulatory compliance.
- Communicated with new component suppliers to achieve advantages in cost and quality.

Embedded Software Engineering Intern , Baykar Technologies – Istanbul, Türkiye	July 2023 – August 2023
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- Prepared and delivered a technical presentation on a communication protocol, including detailed analysis and comparison with alternative communication protocols.

Research and Development Intern , Guleryuz Karoser Otomotiv A.Ş. – Bursa, Türkiye	July 2022 – August 2022
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- Inspected CAN bus scheme placement in an electrical system.
- Standardized signal naming on the model to accelerate production line placement.
- Determined required cable lengths for CAN bus lines using the system model.

Projects

Track Shirt for Tracking Athlete Performance (TUBITAK 1001)

- Designed a PCB in KiCad for textile integration using conductive yarn.
- Programmed the RSL10 chip in C for BLE broadcast-mode communication.
- Implemented acquisition and wireless transmission of acceleration and ECG data.

User Interface and Data Logger for Solar Car

- Developed a C# user interface to receive data via UART and display it in real time.
- Implemented real-time graph visualization using the LiveCharts library.
- Designed an algorithm to verify the accuracy of incoming data. Successfully retrieved and logged 100+ data signals as CSV files in real time for over 1 hour during an actual race without crashes or freezes.

Four-legged Walking Robot Simulation

- Designed and programmed an autonomous quadruped robot, implementing locomotion algorithms in Python using Webots.

Extracurricular Activities

Solar Team Solaris - Electronic Subgroup Member	October 2021 – October 2022
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- Reverse-engineered incoming LoRa data using HTerm to analyze communication frames.
- Designed and developed a C# interface to retrieve and display real-time data from a solar-powered racing car based on the analysis.
- Implemented the CAN bus communication protocol in C on an STM32 microcontroller. Designed and built the required testing and validation circuitry for CAN communication.

Skills

Programming: C, C#, Python, SystemVerilog, MATLAB

Communication Protocols: TCP/IP, UDP, SPI, UART, CAN, I2C, MIL-STD-1553

Software: LTspice, KiCAD, QUCS, Webots

Languages: English (Advanced), Turkish (Native)